

CE 310 — Week 6 Assignment: Empirical Percentiles Across CE Practice

Total: 60 points · Due: Wed Oct 7, 9:00 AM · Submit via D2L

Do not repeat thresholds already used in the in-class exercise (4,000 psi, 65 dB, 65 mph, 85th-percentile speed, \$250/CY, 90th-percentile cost, the AM/PM 85th-percentile comparisons). This assignment uses new thresholds to extend, not duplicate, what you computed in class.

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Where answers go	Column A of the Answers sheet labels every row — P1a, P2a_AM, P2b, and so on. Each question names its row; type into the shaded cell on that row
Column layout	Column A labels the question and Column B says what to find. Enter your formula in Column C and the answer in Column D ; Column E is the unit — do not insert, delete, or move columns
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(D2:D501,">5200")/500, so Excel stores the formula instead of running it. Screenshots are not accepted

Requirement	Detail
Written responses	Type them in the Column D cells for P2b, P2c and P3 on the same sheet — the answer column, same as every other row — do not submit a separate PDF
Name and NetID	Type them in the Name and NetID cells at the top of the Answers sheet — that is what matches your work to your student record. A 5-point deduction applies if the NetID is blank.
File name	CE310_W6_Assignment_<NetID>.xlsx (e.g. CE310_W6_Assignment_abc123.xlsx)
Submission	Upload the .xlsx file to D2L → Assignments by 9:00 AM, Wed Oct 7

Problem 1 — Concrete Strength and Site Acoustics: New Thresholds (30 pts)

P1a–c use column D (fc_psi) of Week5_CE_Measurements.csv, all 500 records unless noted. **P1d** uses column C (NoiseLevel_dB) of Week6_Noise_Measurements.csv.

P1a (8 pts). What percentage of tests exceed 5,200 psi? Week 5’s Normal model estimated this at 4.1% — report your empirical value and, in 1–2 sentences, compare.

P1b (8 pts). What percentage of tests fall strictly between 4,500 and 5,000 psi? Week 5’s Normal model estimated this at 33.9% — report your empirical value and, in 1–2 sentences, compare. (This gap is larger than most you saw in lecture — say why you think that might be, in general terms.)

P1c (7 pts). Compute the empirical **5th percentile** strength (the Eurocode 7 “characteristic value” convention — stricter than ACI 318’s 10th percentile). Week 5’s Normal model estimated this at 3,769 psi. Report your empirical value (exact-rank SMALL **or** interpolated PERCENTILE, your choice — state which) and compare to the model.

P1d (7 pts). What percentage of noise readings exceed 70 dB — a stricter threshold than the exercise’s 65 dB HUD “Acceptable” line? Report your empirical value and, in 1–2 sentences, state what a site-acceptability review would conclude at this stricter threshold versus the exercise’s 65 dB result (18.0%).

Problem 2 — Traffic Speed: New Thresholds and a New Percentile (18 pts)

P2a (6 pts, 2 pts each). What percentage of records show speed above **55 mph** — (i) all records, (ii) AM Peak only, (iii) PM Peak only?

P2b — Written (6 pts). Report the **90th-percentile** speed — a stricter design target than the exercise’s 85th percentile — for (i) all records, (ii) AM Peak, (iii) PM Peak. In 3–5 sentences, explain what moving from the 85th to the 90th percentile does to a posted speed limit, and why a transportation agency might choose one over the other (connect back to the Part 1 debate from lecture).

P2c — Written (6 pts). A colleague argues: “Since Week 5’s Normal model and this week’s empirical values are close for most of these speed questions, we should just always use the Normal model — it’s faster.” In 3–5 sentences, argue for or against this position, citing at least one specific number from Problem 1 or 2 where the empirical and model-based answers diverged meaningfully.

Problem 3 — Reflection (12 pts)

Write a **100–150 word** reflection citing **at least 2 specific numbers** from this assignment or the lecture. Prompt: *A cost estimator on your team says, “AACE guidance is just a suggestion — I always budget to the mean, plus a flat 15% for contingency.” Using what you know about the empirical 90th-percentile cost (\$287.56/CY, or \$274.33/CY for Alpha-only, \$288.86/CY for Beta-only, vs. a mean of \$194.75/CY), argue for or against your colleague’s approach.*

On conventions: \$287.56/CY is your exercise **B10** answer (exact rank, LARGE); your exercise **B11** answer, \$286.50/CY, is the same percentile computed by interpolation (PERCENTILE). The supplier-specific figures above are interpolated. The two conventions differ by about a dollar here, which is far too small to change the argument — cite either, but say which you used.

Grading Summary

Problem	Points
P1 (a+b+c+d)	30
P2 (a + written b + written c)	18
P3	12
Total	60